Project Proposal Form

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GitIntel: Conversational Intelligence for Comprehensive GitHub Repository Analysis

Project Description

Analyzing large GitHub repositories can be challenging due to thousands of commits, hundreds of files, and contributions from multiple developers. It's often difficult to know which parts of the project are changing most, who is contributing what, and how the codebase is evolving over time. **GitIntel** is a smart, chat-based tool that simplifies this process. It extracts structured data like code churn, package-level activity, and developer contributions, and provides clear, human-readable explanations that help users quickly explore and understand even the largest repositories.

Problem Statement & Objectives

Understanding the evolution of large software projects is challenging and time-consuming. Developers, researchers, and project managers often struggle to track changes, identify key contributors, and assess module complexity. Existing tools either provide raw data or require deep technical expertise to interpret.

The main objective of **GitIntel** is to provide an easy-to-use, AI-powered platform that automatically analyzes GitHub repositories and delivers actionable insights in natural language. It aims to help users monitor project activity, understand code evolution, and make informed decisions without manually sifting through commits and files.

Proposed Solution

- **Commit History Analysis:** Use Python tools like PyDriller and GitPython to extract structured data from GitHub repositories.
- **Code Metrics Extraction:** Track code churn (added, deleted, modified lines), package-level activity, and developer contributions.

- **AI-Powered Chat Interface:** Provide human-readable explanations and allow users to ask questions in natural language (Which module changed the most?).
- **Actionable Insights:** Deliver clear, concise summaries that help users understand project evolution without manually browsing commits.
- **Visualization:** Use Neo4j or similar tools to create knowledge graphs for collaboration patterns and module dependencies.
- **Efficient Exploration:** Enable developers, researchers, and project managers to explore large repositories quickly and make informed decisions.

Key Technologies & Tools

Language: Python
Repository Analysis: PyDriller, GitPython
Data Processing & Reporting: Pandas, OpenPyXL
Conversational & AI Integration: Google Generative AI (Gemini)
Knowledge Graph & Visualization: Neo4j
CLI & Utilities: argparse, logging, pathlib, json, datetime
GUI Framework: Tkinter
Version control: Git, Github

Timeline:

	Phase 1				Phase 2				Phase 3			
Task	week 1	week 2	week 3	week 4	week 1	week 2	week 3	week 4	week 1	week 2	week 3	week 4
Prerequisite Knowledge												
Requirements Specification												
Data Collection												
Project Design												
Project Development & Testing												
Final Report Writing												

References:

[1] Samuel Abedu, Laurine Menneron, SayedHassan Khatoonabadi, "RepoChat: An LLM-Powered Chatbot for GitHub Repository Question-Answering," 2025 IEEE/ACM 22nd International Conference on Mining Software Repositories (MSR), Ottawa, ON, Canada, 2025

[2] S. Abedu, A. Abdellatif, and E. Shihab, "LLM-Based Chatbots for Mining Software Repositories: Challenges and Opportunities," EASE '24: Proceedings of the 28th International Conference on Evaluation and Assessment in Software Engineering, 2024

[3] Thomas Berkane, Marie-Laure Charpignon, Maimuna Majumder, “LLM-Based Web Data Collection for Research Dataset Creation” medRxiv

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Signature of the supervisor:

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Proposal Presentation Feedback:

Midterm Presentation Feedback: